RESTful Web Endpoints Quiz (Optimised)

Advanced Architectural Evaluation

Question 01

ARCHITECTURAL STYLE

According to the foundational principles established in 2000, what is the primary definition of REST?

Box A: A specific protocol tied to the HTTP standard for data exchange.

Box B: A transport layer security mechanism for web endpoints.

Box C: An architectural style for distributed systems based on hypermedia.

Box D: A database normalization technique for SQL-based web services.

Box E: A replacement for JSON-based communication in mobile apps.

CORRECT OPTION: C

An architectural style for distributed systems based on hypermedia.

Architectural Rationale

Roy Fielding proposed REST in 2000 as an architectural style for building distributed systems based on hypermedia. While it leverages HTTP, it is theoretically independent of any underlying protocol.

Question 02

STATELESSNESS

In a stateless request model, why is high scalability significantly easier to achieve?

Box A: Because servers store client history in local memory for speed.

Box B: There is no need to retain any affinity between clients and specific servers.

Box C: It forces all requests to be processed by a single master server.

Box D: It allows the backend database to be bypassed entirely.

Box E: It encrypts the payload using session-specific keys automatically.

CORRECT OPTION: B

There is no need to retain any affinity between clients and specific servers.

Architectural Rationale

The stateless constraint means any server can handle any request from any client. Since no transient state is stored between requests on the server, load balancing becomes highly efficient as server affinity is removed.

Question 03

HTTP VERBS

What is the functional difference between the PUT and PATCH HTTP verbs in a RESTful API?

Box A: PUT creates a new record; PATCH deletes an existing one.

Box B: PUT is used for reading data; PATCH is used for writing data.

Box C: PUT replaces an entire resource; PATCH performs a partial update.

Box D: PUT is only for JSON data; PATCH is only for XML data.

Box E: There is no difference; they are functionally interchangeable.

CORRECT OPTION: C

PUT replaces an entire resource; PATCH performs a partial update.

Architectural Rationale

`PUT` either creates or replaces the resource at a specific URI (full replacement). In contrast, `PATCH` applies a set of changes to an existing resource (partial update), reducing bandwidth for large objects.

Question 04

RICHARDSON MATURITY MODEL

At which Level of the Richardson Maturity Model does an API become “Truly RESTful” according to Fielding’s definition?

Box A: Level 0: One URI, all POST requests.

Box B: Level 1: Separate URIs for individual resources.

Box C: Level 2: Use of standard HTTP verbs (GET, POST, etc.).

Box D: Level 3: Use of hypermedia (HATEOAS).

Box E: Level 4: Automated service evolution.

CORRECT OPTION: D

Level 3: Use of hypermedia (HATEOAS).

Architectural Rationale

Level 3 corresponds to a truly RESTful API. It utilizes HATEOAS, allowing clients to discover all available actions and related resources via links provided within the response body itself.

Question 05

URI NAMING STRATEGY

Which of the following URI designs adheres to REST best practices for resource organization?

Box A: `/create-order`

Box B: `/orders/delete/5`

Box C: `/orders/5`

Box D: `"/updateCustomerInfo"`

Box E: `"/all-orders-list"`

CORRECT OPTION: C

/orders/5

Architectural Rationale

REST URIs should be based on nouns (resources) rather than verbs (operations). `/orders/5` identifies a specific item in a collection. Operations like creating or deleting should be handled by HTTP verbs (POST or DELETE) on that URI, not by embedding verbs in the path.

Question 06

ASYNCHRONOUS OPS

When a REST operation requires significant processing time, which HTTP status code and header combination is recommended?

Box A: 200 OK with the result body.

Box B: 404 Not Found with an Error header.

Box C: 202 Accepted with a Location header for status polling.

Box D: 503 Service Unavailable with a Retry-After header.

Box E: 301 Moved Permanently with a New-URI header.

CORRECT OPTION: C

202 Accepted with a Location header for status polling.

Architectural Rationale

For long-running tasks, returning `202 Accepted` prevents client timeouts. The `Location` header directs the client to a status endpoint where they can poll for the progress or eventual result of the operation.

Question 07

LEVEL 1 MATURITY

What is the defining characteristic of Level 1 in the Richardson Maturity Model?

Box A: Using a single endpoint for all XML traffic.

Box B: Implementation of separate URIs for individual resources.

Box C: Mandatory use of JSON over XML.

Box D: Support for cross-origin resource sharing (CORS).

Box E: Automatic encryption of resource representations.

CORRECT OPTION: B

Implementation of separate URIs for individual resources.

Architectural Rationale

Level 1 introduces the concept of individual URIs for specific resources resources (e.g., /orders/1, /orders/2) rather than having everything go through a single root endpoint as seen in Level 0.

Question 08

DATA ABSTRACTION

Why should a web API avoid simply mirroring the internal structure of a relational database?

Box A: Relational databases are too slow for web traffic.

Box B: To isolate clients from changes to the underlying database schema.

Box C: Because JSON cannot represent table relationships.

Box D: REST requires every table to be merged into one.

Box E: It would violate the HTTP protocol's internal security rules.

CORRECT OPTION: B

To isolate clients from changes to the underlying database schema.

Architectural Rationale

A well-designed API acts as an abstraction layer. By introducing a mapping layer, the business entities exposed to the client can evolve independently of the physical database schema, ensuring stability for existing clients.

Question 09

LEVEL 2 MATURITY

In practice, many published web APIs fall around Level 2. What does Level 2 specifically introduce?

Box A: OAuth2 authentication.

Box B: Gzip compression for all responses.

Box C: Using HTTP methods (GET, POST, etc.) to define operations.

Box D: Multi-language localization for error messages.

Box E: Server-side rendering of resource data.

CORRECT OPTION: C

Using HTTP methods (GET, POST, etc.) to define operations.

Architectural Rationale

Level 2 is characterized by the use of the full range of standard HTTP verbs to perform actions on resources. This standardizes how clients interact with services (e.g., DELETE to remove, POST to create).

Question 10

CHATTY APIS

What is a primary disadvantage of a ‘chatty’ web API that exposes many small resources?

Box A: It makes the database too large.

Box B: It requires multiple requests to find all required data, increasing load and latency.

Box C: It prevents the use of HTTPS encryption.

Box D: Clients are unable to parse JSON responses.

Box E: It forces the API to use XML instead of JSON.

CORRECT OPTION: B

It requires multiple requests to find all required data, increasing load and latency.

Architectural Rationale

Chatty APIs increase the load on the server and the latency for the client because several round-trips are needed to fetch related information. Denormalizing data into larger resources can mitigate this overhead.